House Prediction Analysis Report

Title

Regression Analysis for House Price Prediction Using Simple Linear Regression

Student Name: Blessing Ogar

Internship Task

Level 2 (Intermediate) – Task 1: Regression Analysis

1. Introduction

Machine Learning has become an important tool for predicting future outcomes based on historical data. One of the most commonly used predictive techniques is Linear Regression, which models the relationship between a dependent variable and one or more independent variables.

This internship task focused on implementing a Simple Linear Regression model to predict house prices based on housing characteristics using Python and Scikit-Learn.

2. Objectives

The objectives of this task were to:

- Perform a simple linear regression analysis.
- Split the dataset into training and testing sets.
- Train a regression model using Scikit-Learn.
- Predict house prices using the trained model.
- Evaluate model performance using Mean Squared Error and R-Squared.
- Interpret regression coefficients.

3. Tools and Technologies Used

The following tools were used during the implementation:

- Python Programming Language
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-Learn

4. Dataset Description

The House Prediction Dataset was used for this analysis. The dataset contains 506 records and 14 attributes related to housing conditions and prices.

For this task:

- Predictor Variable (X): RM (Average Number of Rooms)
- Target Variable (Y): MEDV (Median House Value)

5. Methodology

5.1 Data Loading

The dataset was loaded into a Pandas DataFrame and inspected for consistency.

5.2 Data Exploration

The following exploratory checks were performed:

- Dataset dimensions
- Data types
- Summary statistics
- Missing value analysis

5.3 Feature Selection

RM was selected as the independent variable because it has a strong relationship with house prices.

MEDV was selected as the dependent variable representing house values.

5.4 Data Splitting

The dataset was divided into:

- Training Set (80%)
- Testing Set (20%)

This was achieved using the train_test_split() function from Scikit-Learn.

5.5 Model Training

A Linear Regression model was created and trained using the training data.

5.6 Prediction

The trained model was used to predict house prices on the testing dataset.

5.7 Evaluation

Model performance was evaluated using:

- Mean Squared Error (MSE)
- R-Squared (R^2)

6. Results and Discussion

The model produced a positive regression coefficient, indicating that house prices increase as the number of rooms increases.

The regression line demonstrated a positive linear relationship between the selected variables.

The R-Squared value showed the proportion of house price variation explained by the number of rooms.

The Mean Squared Error measured the prediction error of the model.

Overall, the model successfully demonstrated the application of Simple Linear Regression for predictive analysis.

7. Challenges Encountered

During implementation, several challenges were encountered:

- Installation of Scikit-Learn package.
- File path errors while loading the dataset.
- Understanding the interpretation of regression coefficients.

These challenges were resolved through package installation, path verification, and additional study of regression concepts.

8. Conclusion

This internship task provided practical experience in implementing Simple Linear Regression using Python. The project demonstrated how machine learning can be used to predict house prices based on housing characteristics.

The exercise improved skills in data preprocessing, model development, evaluation, and interpretation of machine learning models.

9. Recommendations

Future work can improve prediction accuracy by:

- Using multiple predictor variables.
- Applying Multiple Linear Regression.
- Performing feature engineering.
- Comparing different machine learning algorithms.

References

Scikit-Learn Documentation

Pandas Documentation

Matplotlib Documentation

Python Software Foundation Documentation